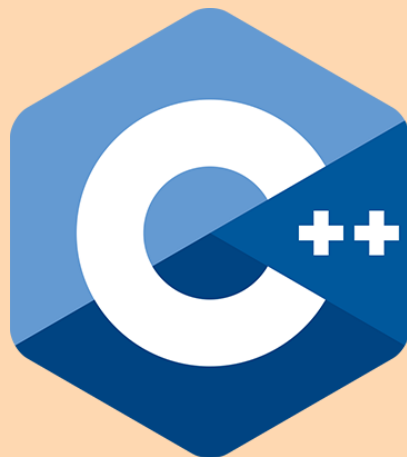


3. WHILE . . .

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Sep. 24, 2026

Review

If Statement

Switch Statement

Logical Expressions

Overview

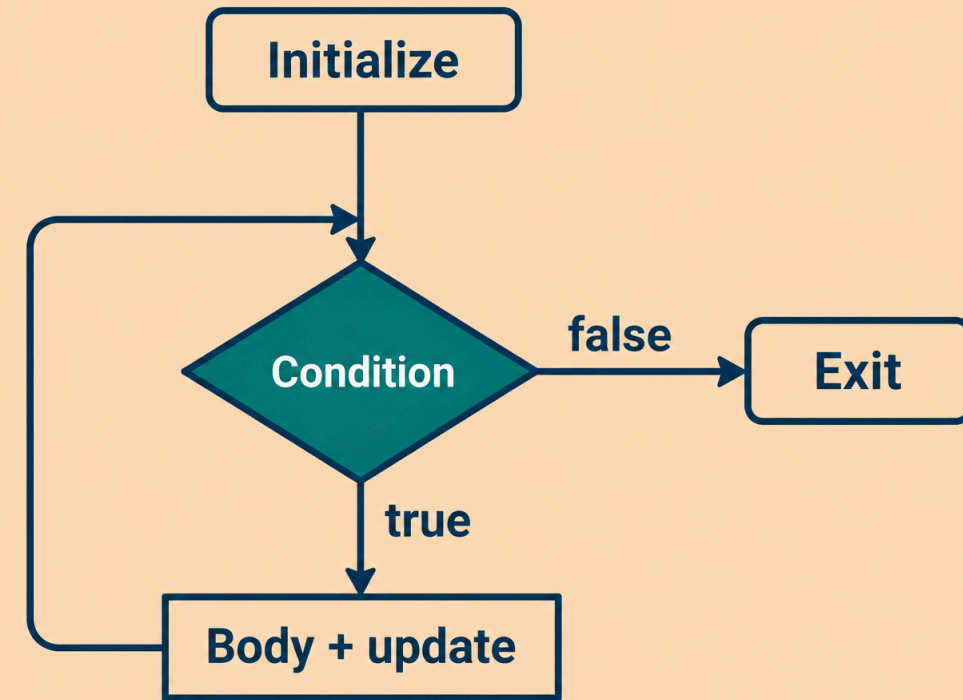
While (Do-While) Statement

break Statement

while Statement

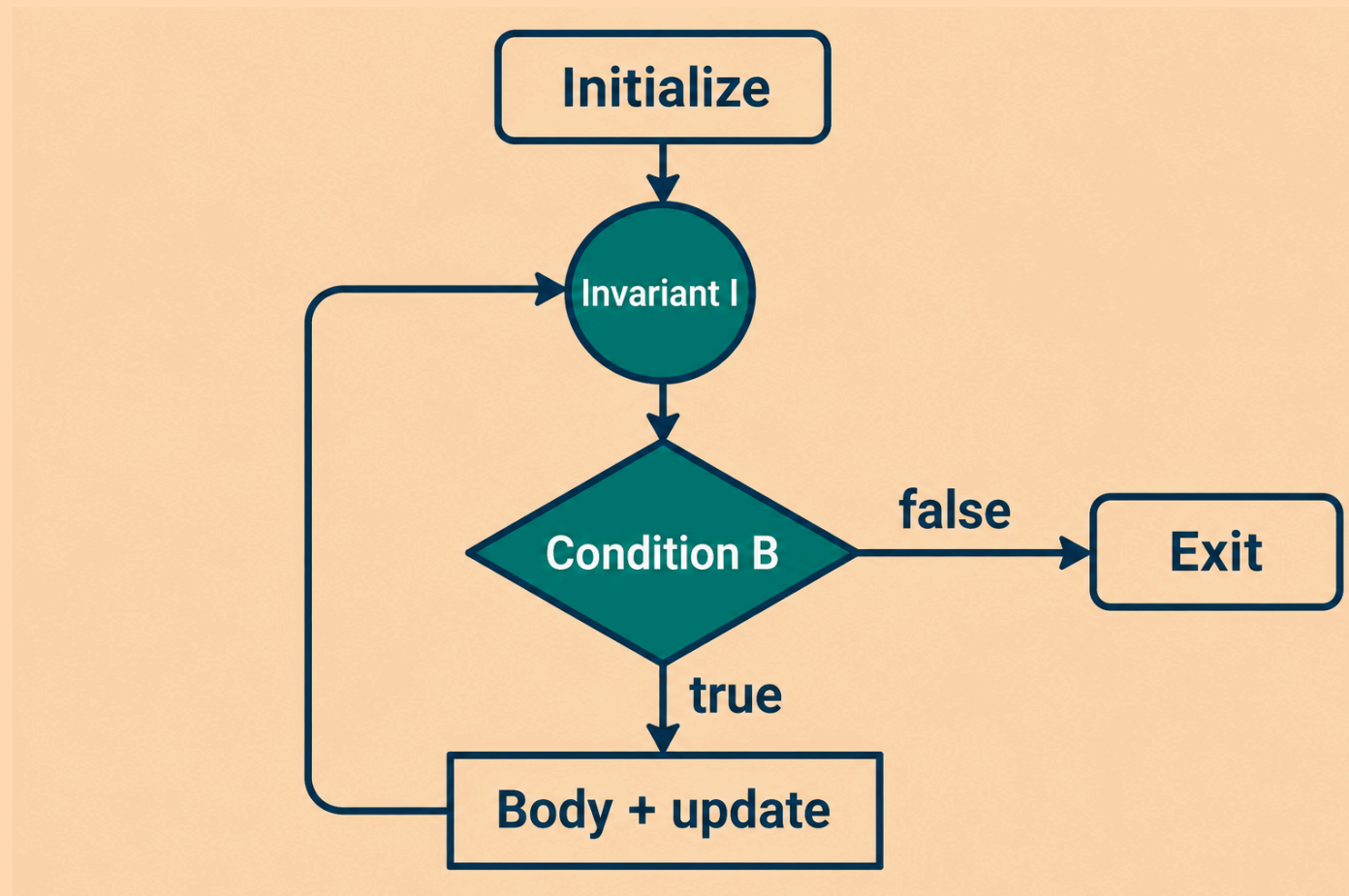
```
// initialize  
while (condition) {  
    // body + update  
}
```

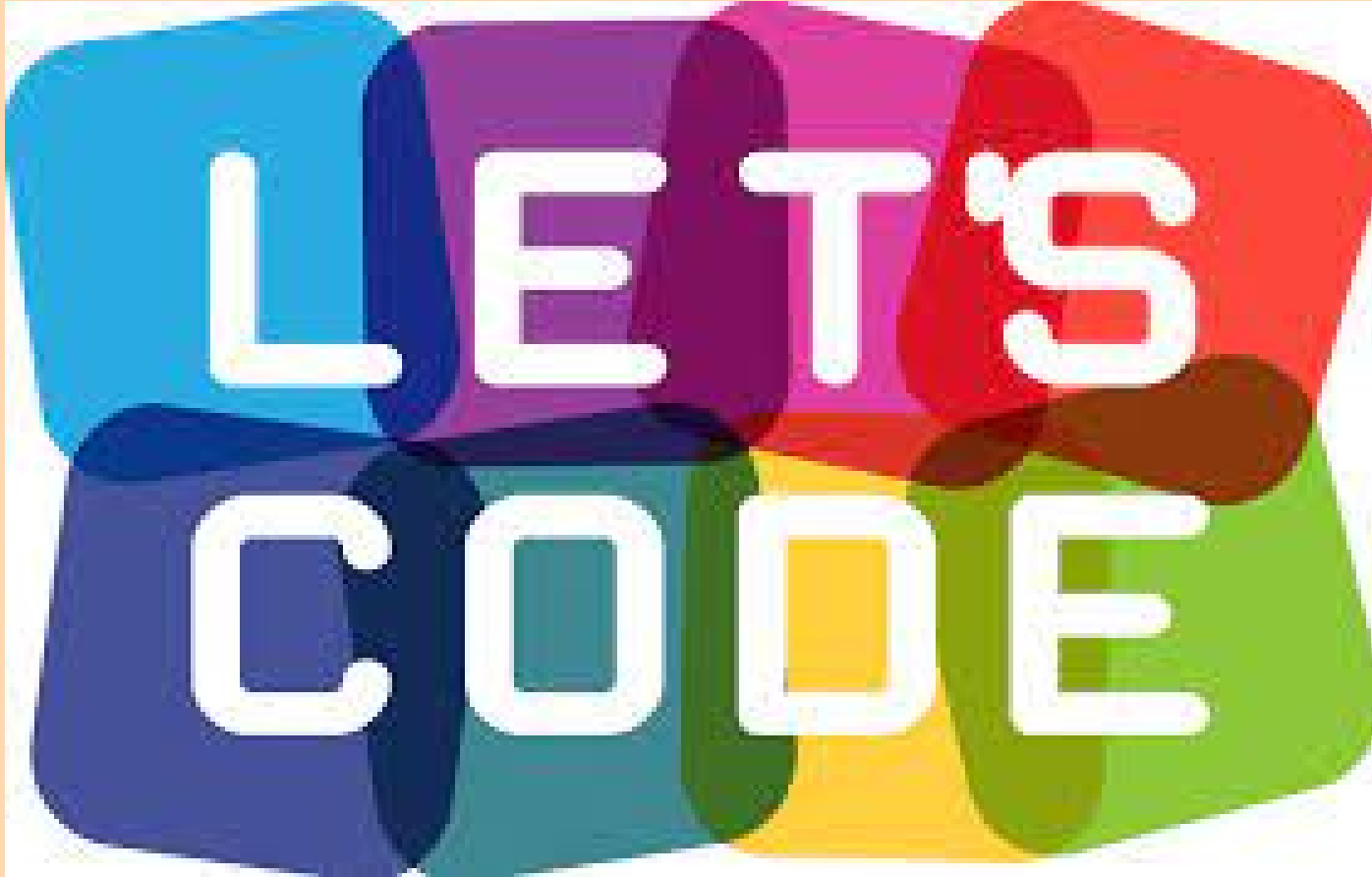
May execute **zero** times.



Loop Invariant as Abstraction

What remains true before every iteration?





euclid.cpp fib.cpp digits.cpp
palindrome.cpp binary-search.cpp

Greatest Common Divisor

$$\gcd(a, b) = \gcd(b, a \% b)$$



$$(48, 18) \longrightarrow (18, 12) \longrightarrow (12, 6) \longrightarrow (6, 0)$$

Loop Invariant

$a_0, b_0 > 0$: **original inputs**

$$\boxed{\gcd(a, b) = \gcd(a_0, b_0)}$$

$$(a, b) \xrightarrow{b \neq 0} (b, a \% b)$$

Fibonacci Sequence

$$F_0 = 0$$

$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \quad (n > 1)$$

Loop Invariant

i : index, $0 \leq i \leq n$

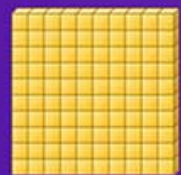
$\text{previous} = F_i$	$\text{current} = F_{i+1}$
-------------------------	----------------------------

(F_i, F_{i+1})	$\longrightarrow$	$(F_{i+1}, F_i + F_{i+1})$
$\underbrace{\hspace{1.5cm}}$		$\underbrace{\hspace{1.5cm}}$
before		after: (F_{i+1}, F_{i+2})

Number of Digits

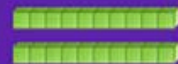


Identify the Place & Value
of Digits in a
Three-Digit Number



1 Hundred

127

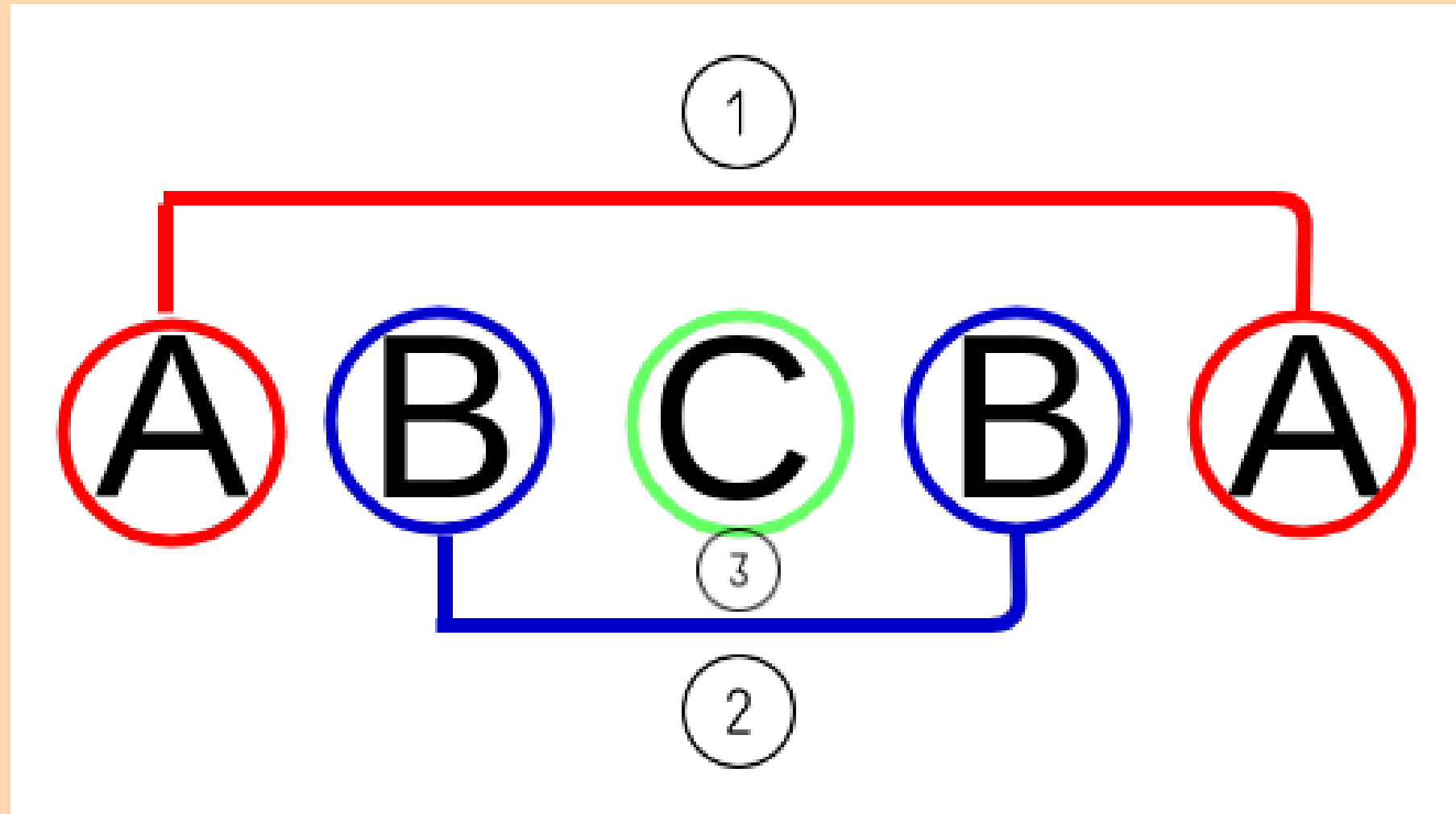


2 Tens



7 Ones

Palindrome



Binary Search

